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BHARATIYA ANTARIKSH HACKATHON 2025

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Team Name: **Pavitra**

Team Leader Name: Evani Menon

Problem Statement: **Developing an Algorithm for Air Quality Visualizer and Forecast App to Generate Granular, Real-time, and Predictive Air Quality Information**

Team Members

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Brief about the Idea:

Most existing Air Quality Index (AQI) tools focus only on large cities, **leaving small towns and rural regions underserved**. Our idea is to build an intelligent **Air Quality Visualizer and Forecast App** that delivers granular, real-time, and predictive air pollution insights for **any location** in India, including tier 2 & 3 towns and remote areas.

Unlike existing tools focused on metros, **we integrate multiple open data sources** (e.g., ISRO, CPCB, IMD) to fill data gaps in underserved regions. By combining spatial, temporal, and meteorological data, we provide **intuitive heatmaps, forecasts, and historical trends in a single interface**. Our unique selling point lies in delivering high-resolution, **rural-inclusive air monitoring powered by predictive modeling and a proactive notification system**, making air quality awareness accessible and actionable nationwide.

Unique Selling Point of our Algorithm:

To **overcome data sparsity** in rural areas and small towns, we integrate ground-based data from **CPCB and IMD**, satellite-derived data from **ISRO Bhuvan** as well as **weather and geospatial patterns** to enhance spatial coverage.

Our model is built on the **CatBoost gradient boosting framework**. It's known for its high performance on large, structured environmental datasets. CatBoost efficiently handles the diverse **feature set of pollutants** (PM2.5, PM10, NO, NO_2, NOx, SO_2, CO, O_3, NH_3, Benzene, Toluene, Xylene 10) and **meteorological factors** (Temperature, Humidity, Wind Speed/Direction, Solar Radiation, Barometric Pressure, Rainfall, etc.), providing accurate, interpretable AQI forecasts without the complexity or training time of deep learning models. This model structure is inspired by recent peer-reviewed papers in atmospheric forecasting and environmental ML.

List of Features:

- **Real-time AQI visualization** using live location and colour coded zones
- **Health-based recommendations** and **push notifications** that display messages such as “Wear a mask”, and suggest indoor vs outdoor activity levels
- **AQI forecasts** for the next 24-72 hours using machine learning
- **Historical trends** over the past few weeks or months
- **AQI pattern maps** displaying how AQI changes in each season
- **Interactive pollution heatmaps**
- **Citizen reporting/ Feedback loop** by letting users report pollution events (like burning or factory smoke)
- **Wearable device integration** to sync with smartwatches

Process Flow Diagram:

```
[User]
   |
   | 1. Opens app or website
   v
[Frontend Interface: Mobile/Web]
   |
   | 2. Sends location or pin
   v
[Backend Server]
   |
   | 3. Fetches real-time AQI
   |    ↳ [CPCB API / WAQI API]
   |    ↳ [ISRO Bhuvan (Satellite Layer)]
   |    ↳ [OpenWeatherMap API]
   |
   | 4. Retrieves historical data
   |    ↳ [CPCB / data.gov.in]
   |
   | 5. Sends combined data to:
   v
   .
```

```
   .
   v
[ML Forecasting Engine]
   |  [] CatBoost model trained on pollutants + weather
   |  [] Output: AQI forecast for next 24–72 hrs
   v
[Database]
   |  [] Store historical AQI, forecast, user reports
   |
   v
[Alert Engine / Push Notification System]
   |
   | 6. Sends alerts based on thresholds:
   |     - “Wear a mask”
   |     - “Avoid outdoor activity”
   v
[User]

(Loop continues every time data updates or app is opened)
```

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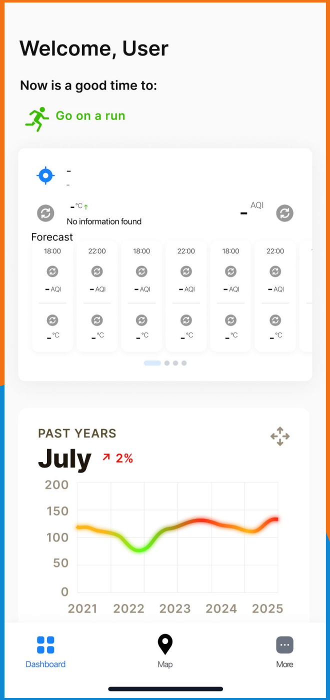
Wireframes:

Pavitra
😷 Pollution Spike Nearby! Crop burning is raising pollution near your area. AQI: 215. Close windows & avoid outdoor travel!

Custom push notifications and activity suggestions

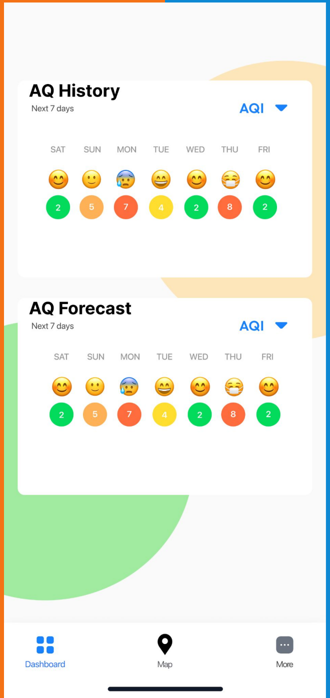


Personalised health recommendations

Live AQI and weather throughout the day

AQI History and Forecast

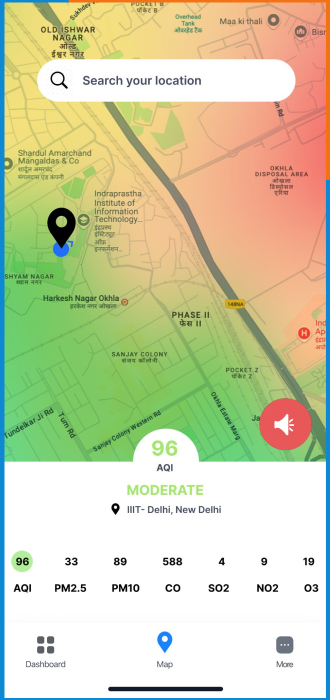
AQI Trends



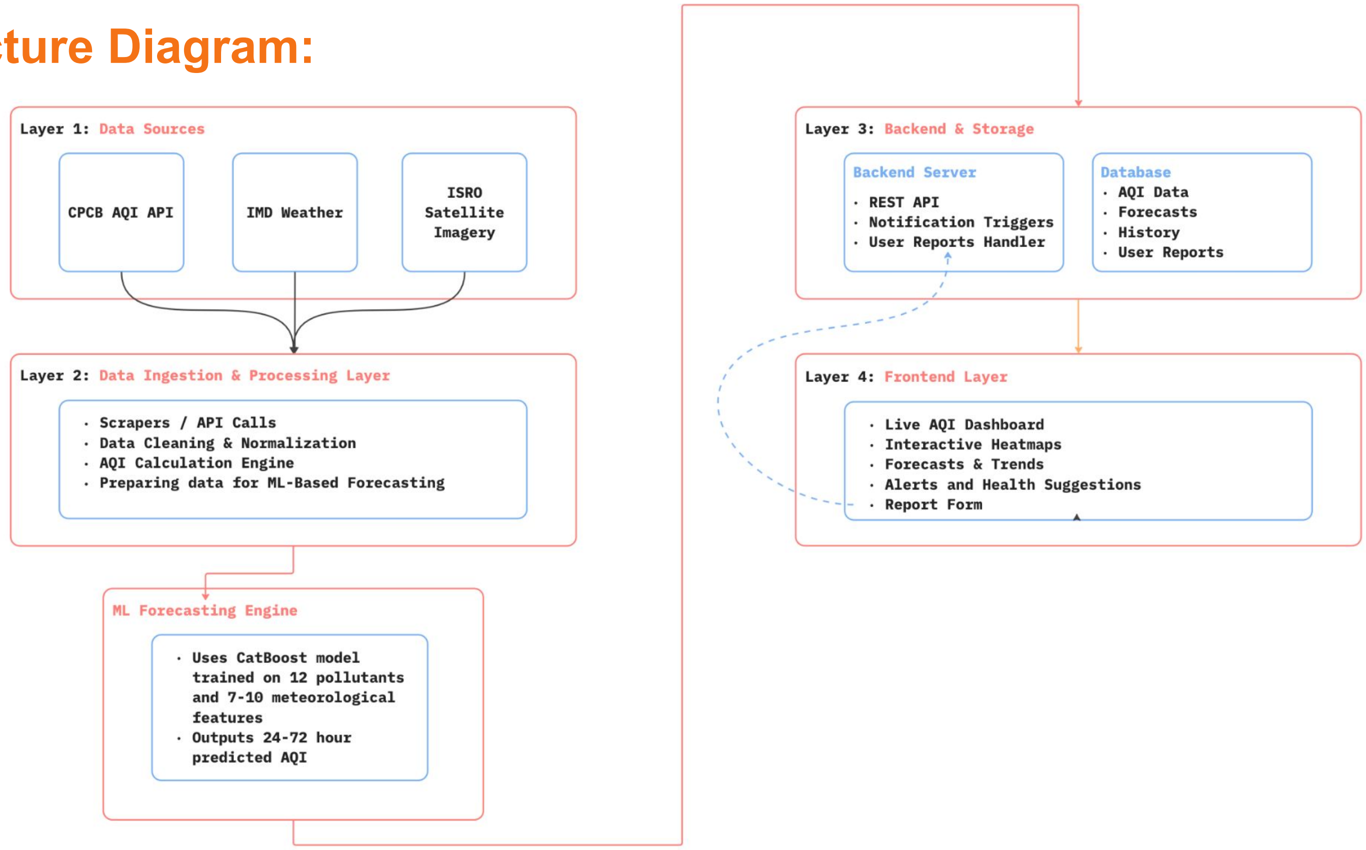
Real-time visualisation using live location and heatmaps

Citizen reporting/ feedback button

Wearable device integration for alerts



Architecture Diagram:



Technologies to be Used

Category	Technology/Tool	Purpose/ Usage
Frontend	Flutter (or React Native)	Cross-platform mobile app for Android/iOS
	React.js	Web dashboard or quick MVP prototyping
	Leaflet.js / Google Maps JS API	Interactive map with AQI overlays, markers, heatmaps
	Chart.js / Plotly.js	AQI trend charts and forecast graph visualization
Backend	Node.js + Express.js	Lightweight server for API handling and ML integration (optional for MVP)
	Supabase / Firebase Realtime DB	Backend-as-a-service for storing reports, user settings, auth

APIs/ Data Sources	WAQI (World Air Quality Index)	Real-time AQI data and pollutant breakdown
	OpenWeatherMap API	Weather parameters (wind, humidity, temperature) for ML input
	ISRO Bhuvan / Bhoonidhi	Satellite-based pollution layers (AOD, NO_2, fire data)
	CPCB Portal / data.gov.in	Historical AQI data from Indian ground stations for ML training
Machine Learning	Python + Pandas / Scikit-learn	Data preprocessing, model training
	Google Colab / Kaggle Notebooks	Free cloud GPU environment for training models
	TensorFlowLite	Run trained models efficiently on mobile devices
Deployment	Firebase Hosting	Host mobile backend or React web dashboard
	Vercel	Rapid hosting for the frontend MVP
Notifications	Firebase Cloud Messaging	Push alerts when AQI crosses thresholds
Analytics	Firebase Analytics / Google Analytics	Track user activity and feature usage

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